

CLL371 Project 2024-25

Ammonia Production Using Hydrogen and Nitrogen



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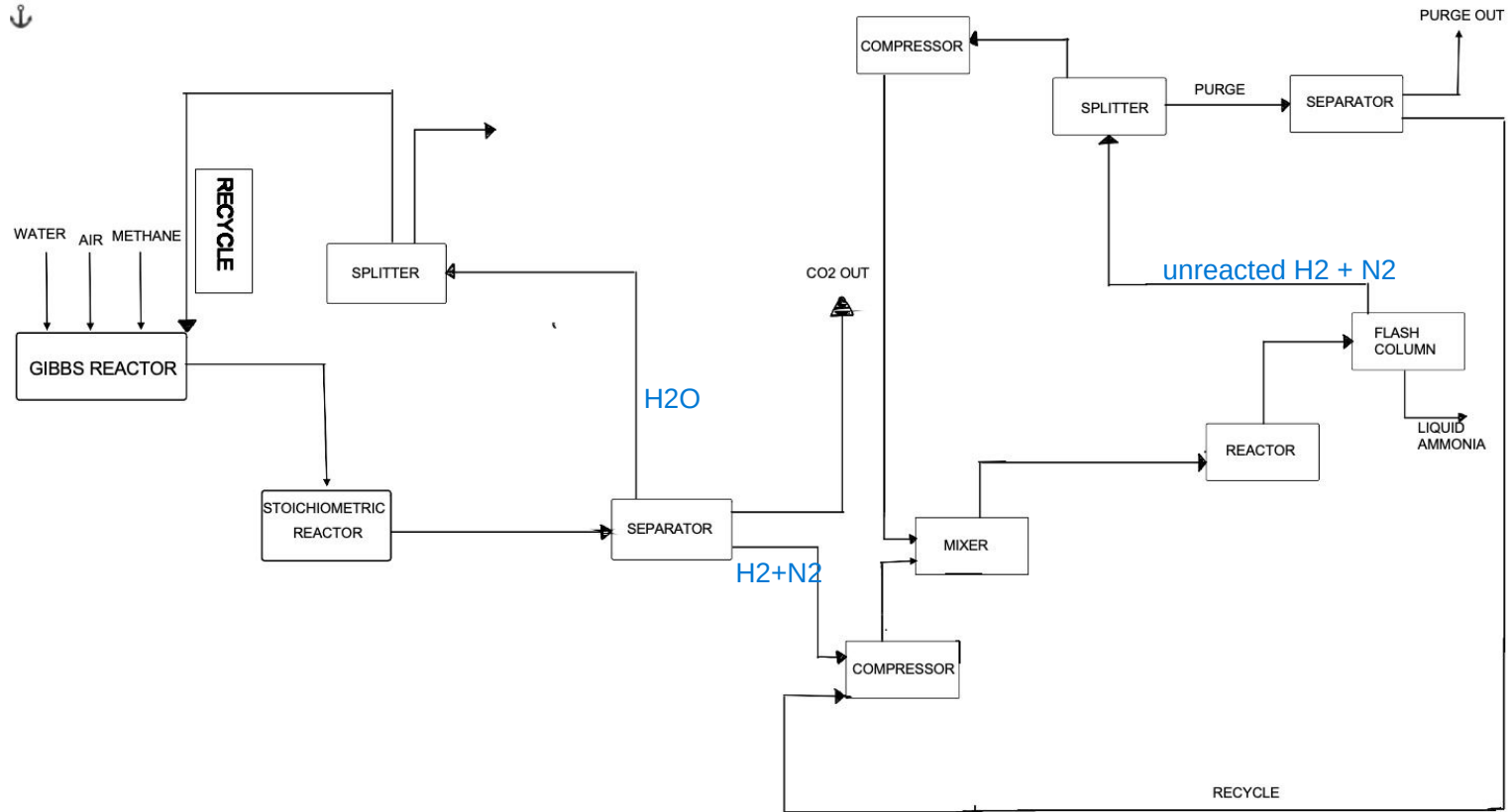
NIKHIL GUPTA, 2022CH11016

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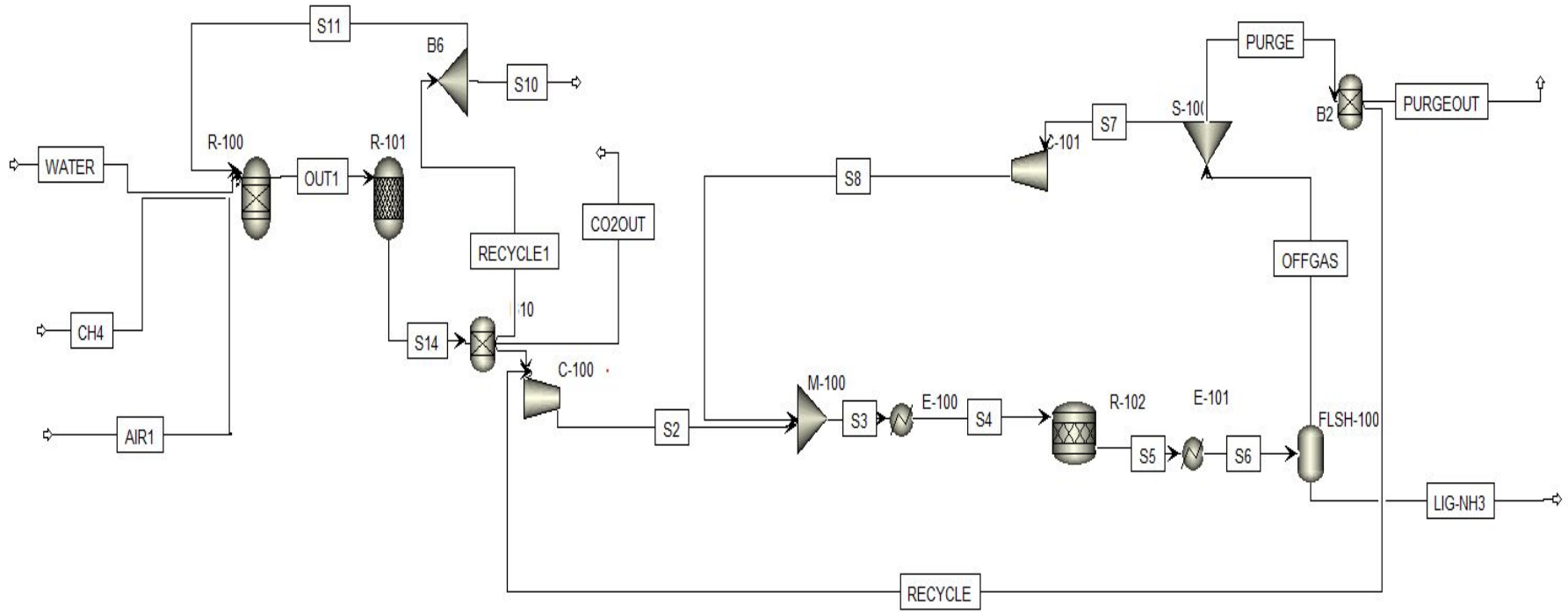
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SNIGDHADYUT DASH, 2022CH71482

BLOCK FLOW DIAGRAM



PROCESS FLOW DIAGRAM



REACTIONS INVOLVED

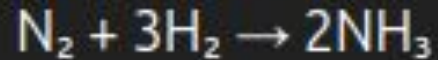
Steam Methane Reforming (SMR)



Water-Gas Shift Reaction



Ammonia Synthesis



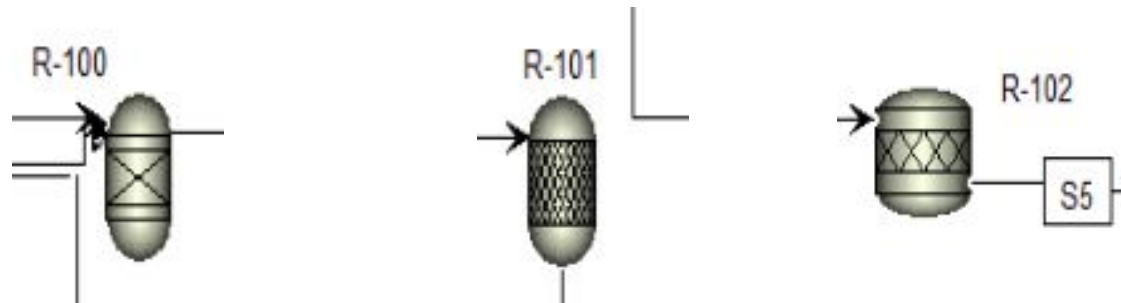
STREAM DATA

MIXED PHASE

Mixed phase											
Units	LIG-NH3	OFFGAS	OUT1	PURGE	PURGEOUT	RECYCLE	RECYCLE1	S1	S2	S3	
From	FLSH-100	FLSH-100	B1	S-100	B2	B2	B10	B10	C-100	M-100	
To		S-100	B7	B2		M-100	B6	C-100	M-100	E-100	
Phase		Liquid Phase	Vapor Phase	Vapor Phase	Vapor Phase	Vapor Phase	Vapor Phase	Vapor Phase	Vapor Phase	Vapor Phase	
Temperature	K	300.15	300.15	1123.15	300.15	280.8627	280.8627	623.1491	623.1491	1419.326	966.6893
Pressure	atm	270	270	24.67308	270	1	1	24.67308	24.67308	271.4	1
Molar Vapor Fraction		0	1	1	1	1	1	1	1	1	1
Molar Liquid Fraction		1	0	0	0	0	0	0	0	0	0
Mole Flows	kmol/hr	5578.593	7644.919	31120.04	76.44919	0.293178	76.15363	17558.46	11046.65	11046.65	18691.04
H2	kmol/hr	49.04541	2931.785	7572.745	29.31785	0.293178	29.02449	0	8250.649	8250.649	11182.12
N2	kmol/hr	62.05577	4038.591	2796.004	40.38591	0	40.38406	0	2796.004	2796.004	6834.411
CH4	kmol/hr	0	0	21.16672	0	0	0	20.95505	0	0	0
AR	kmol/hr	0	0	22	0	0	0	0	0	0	0
CO	kmol/hr	0	0	713.5834	0	0	0	0	0	0	0
NH3	kmol/hr	5467.492	674.5436	7.991111	6.745436	0	6.745069	0	0	0	674.5069
H2O	kmol/hr	0	0	18392.56	0	0	0	17537.51	0	0	0
O2	kmol/hr	0	0	3.86E-14	0	0	0	0	0	0	0
CO2	kmol/hr	0	0	1593.995	0	0	0	0	0	0	0
Volume Flow	l/min	3248.894	13301.34	1943663	133.0134	112.6853	29253.2	579013.2	385224.8	82756.17	24717101
Units	S4	S5	S6	S7	S8	S10	S11	S14	WATER		
From	E-100	R-100	E-101	S-100	C-101	B6	B6	B7			
To	R-100	E-101	FLSH-100	C-101	M-100		B1	B10	B1		
Phase		Vapor Phase	Vapor Phase		Vapor Phase	Vapor Phase	Vapor Phase	Vapor Phase	Vapor Phase		
Temperature	K	755	755	300.15	300.15	300.5722	623.1491	623.1491	623.15	823	
Pressure	atm	1	270	270	270	271	24.67308	24.67308	24.67308	24.67308	
Molar Vapor Fraction		1	1	0.578131	1	1	1	1	1	1	
Molar Liquid Fraction		0	0	0.421869	0	0	0	0	0	0	
Mole Flows	kmol/hr	18691.04	13223.51	13223.51	7568.47	7568.233	175.5846	17382.88	31120.04	4000	
H2	kmol/hr	11182.12	2980.83	2980.83	2902.467	2902.449	0	0	8250.649	0	
N2	kmol/hr	6834.411	4100.646	4100.646	3998.205	3998.022	0	0	2796.004	0	
CH4	kmol/hr	0	0	0	0	0	0.209551	20.7455	21.16672	0	
AR	kmol/hr	0	0	0	0	0	0	0	22	0	
CO	kmol/hr	0	0	0	0	0	0	0	35.67917	0	
NH3	kmol/hr	674.5069	6142.036	6142.036	667.7981	667.7618	0	0	7.991111	0	
H2O	kmol/hr	0	0	0	0	0	175.3751	17362.13	17714.65	4000	
O2	kmol/hr	0	0	0	0	0	0	0	3.86E-14	0	
CO2	kmol/hr	0	0	0	0	0	0	0	2271.9	0	
Volume Flow	l/min	19305832	54693.3	16550.23	13168.32	13147.55	5790.132	573223	1063457	179795.7	

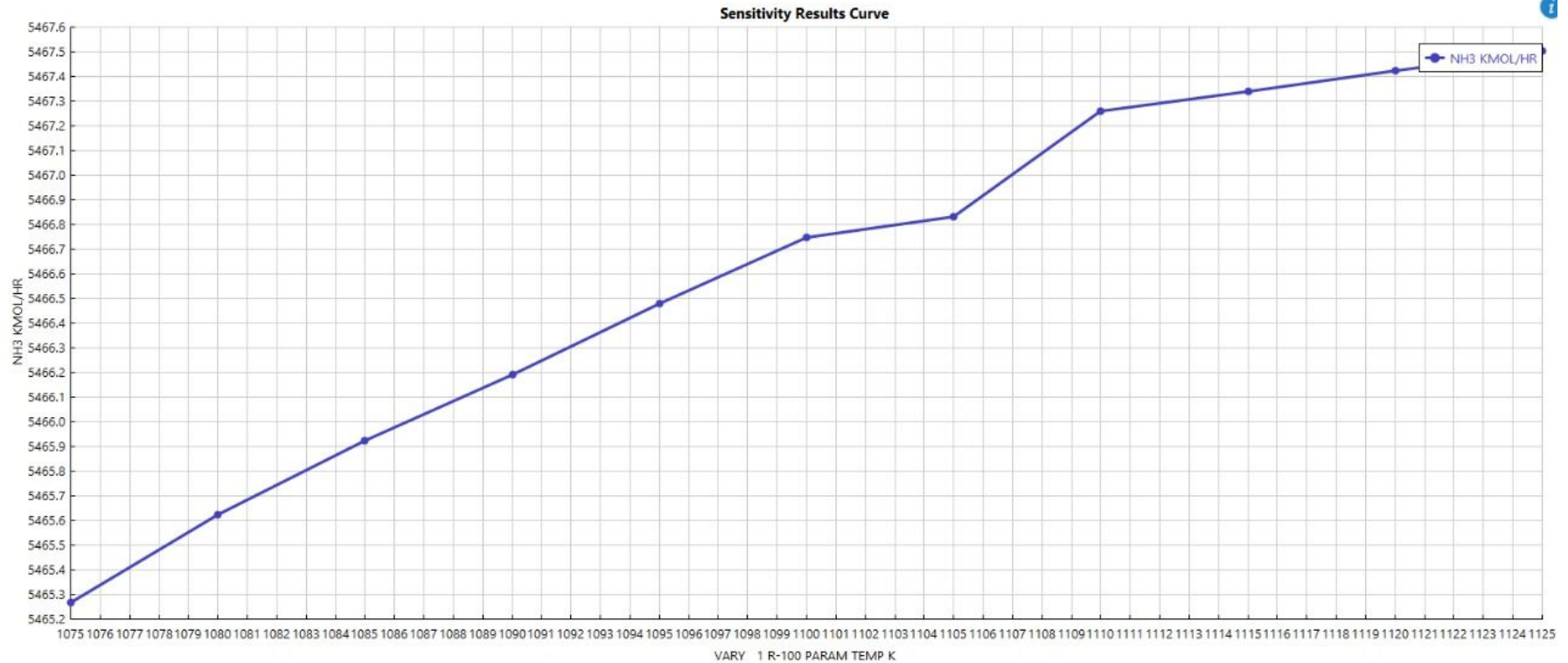
REACTOR DATA

	R-100	R-101	R-102
Reactor Type	Gibbs Reactor	Stoichiometric reactor	Stoichiometric reactor
Reaction	Steam Methane reforming	Water Gas Shift	Ammonia Synthesis
Pressure	25 bar	25 bar	273.577 bar
Temperature	1123K	623K	755K



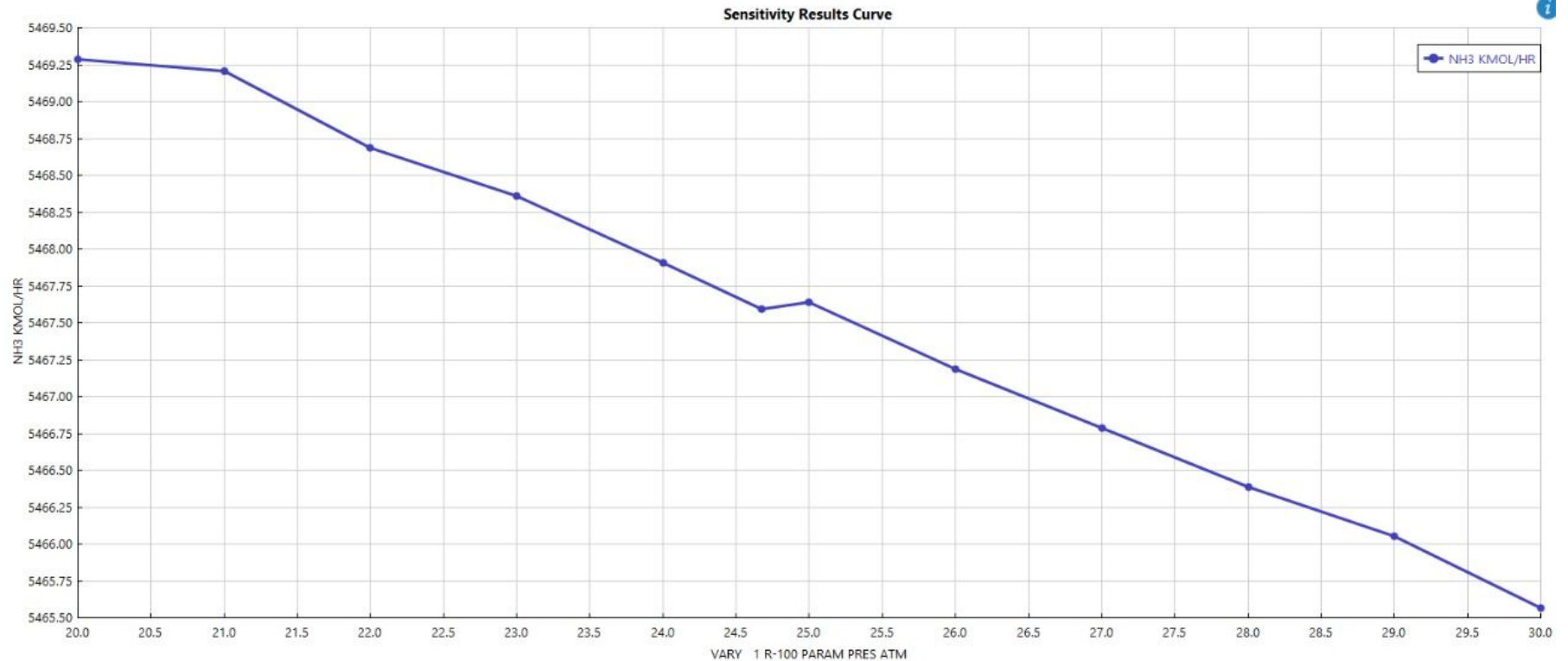
SENSITIVITY ANALYSIS

TEMPERATURE VARIATION



SENSITIVITY ANALYSIS

PRESSURE VARIATION



SIMULATION RESULTS

Parameter	Reformer (B7)	CO ₂ Removal (C-100)	Compressor (M-100)	Reactor (R-100)	Flash Drum (FLSH-100)
Temperature	850 °C	40 °C	45 °C	200 °C	35 °C
Pressure	250 kPa	240 kPa	3000 kPa	2900 kPa	2800 kPa
Molar Flow Rate	800 kmol/h	700 kmol/h	690 kmol/h	685 kmol/h	150 kmol/h (NH ₃ product)
Main Outlet Component	H ₂ , CO, CO ₂	H ₂ , CO (less CO ₂)	H ₂ , N ₂ , CO	NH ₃ , H ₂ , N ₂	NH ₃
CO ₂ Recovery	–	~90%	–	–	–
CH ₄ Conversion	~80%	–	–	~99% (N ₂ + H ₂ → NH ₃)	–



THANK YOU